

# MAXIMILIAN LARTER

I am a plant ecophysiologicalist and an evolutionary biologist.

My primary research interest is understanding how plants respond and adapt to their environment, especially under climate change. I study functional traits and key physiological processes, and use statistics and models to predict the impacts of climate change on the performance, survival, and future distributions of wild and cultivated species.



## RESEARCH EXPERIENCE

- 2025 • Postdoc - modelling urban heat islands and carbon storage  
Sylvain Delzon's lab INRAE, Bordeaux, France  
• Model the impacts of management, development and land use changes on local temperature and carbon storage on the university campus.
- 2024 | 2021 • Postdoc - Plant hydraulics & trait trade-offs, forest ecology and biogeography  
Sylvain Delzon's lab INRAE, Bordeaux, France
- 2021 | 2019 • Postdoc - Herb hydraulics, positive root pressure and drought resistance in Brassicaceae  
Frederic Lens's lab Naturalis, Leiden, The Netherlands
- 2019 | 2017 • Postdoc - Evolution of the anthocyanin pathway in Iochrominae  
Stacey Smith's lab University of Colorado, Boulder, USA
- 2016 | 2012 • PhD - Evolution of cavitation resistance in conifers  
Sylvain Delzon's lab Université de Bordeaux, France  
• Supervisors: Sylvain Delzon and Jean-Christophe Domec



## EDUCATION

- 2025 | 2022 • Qualification CNU  
Sections 66 Physiologie & 67 Biol. des populations et écologie
- 2016 | 2012 • PhD in evolutionary, functional and community ecology.  
Thesis: Evolution of cavitation resistance in conifers  
Université de Bordeaux Bordeaux, France  
• Supervisors: Sylvain Delzon and Jean-Christophe Domec
- 2011 • MSc - Terrestrial Ecosystem Functioning and Modelling  
Université de Bordeaux Bordeaux, France
- 2008 • BSc - Organismal Biology  
Université d'Orléans Orléans, France



## SELECTED PUBLICATIONS

- 2025 • M Larter, G Charrier, S Delzon, W Hammond, A Baranger, C Bertrand, N Martin-StPaul, G Kunstler (2025) Weak global trade-off between frost and drought resistance in trees, *New Phytologist*. [pdf](#)
- 2024 • A Baranger, T Cordonnier, G Charrier, S Delzon, M Larter, N Martin-StPaul, G Kunstler (2024) Living on the edge - physiological tolerance to frost and drought explains range limits of 35 European tree species, *Ecography*. [pdf](#)

## CONTACT

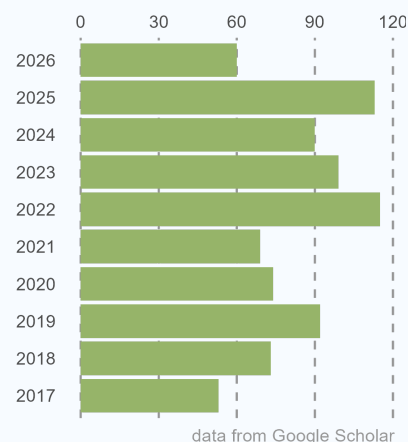
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33600 Pessac, France
- August 5th, 1987. Derby (UK)

## SKILLS

- Native speaker in English and French
- data analysis, statistics (SAS, R, git), modelling
- QGIS, Inkscape
- Plant hydraulics, physiology, gas exchange
- Molecular biology, phylogenetics

## CITATIONS

873 citations, h-index 12



[full publication list](#)

Last updated on 2026-05-18.

- 2019 ● M Larter, A Dunbar-Wallis, AE Berardi, SD Smith (2019) Developmental control of convergent floral pigmentation across evolutionary timescales, *Developmental dynamics*, 248 (11), 1091-1100. [pdf](#)
- 2018 ● M Larter, A Dunbar-Wallis, AE Berardi, SD Smith (2018) Convergent evolution at the pathway level: predictable regulatory changes during flower color transitions, *Molecular biology and evolution*, 35 (9), 2159-2169. [pdf](#)
- 2017 ● M Larter, S Pfautsch, JC Domec, S Trueba, N Nagalingum, S Delzon (2017) Aridity drove the evolution of extreme embolism resistance and the radiation of conifer genus *Callitris*, *New Phytologist*, 215 (1), 97-112. [pdf](#)
- 2015 ● M Larter, TJ Brodribb, S Pfautsch, R Burlett, H Cochard, S Delzon (2015) Extreme aridity pushes trees to their physical limits, *Plant Physiology*, 168 (3), 804-807. [pdf](#)
- 2014 ● PS Bouche, M Larter, JC Domec, R Burlett, P Gasson, S Jansen, S Delzon (2014) A broad survey of hydraulic and mechanical safety in the xylem of conifers, *Journal of Experimental Botany*, 65 (15), 4419-4431. [pdf](#)



## GRANTS AND FUNDING

- Active participation in grant writing  
ANR projects, Calls for Recherche Inter-disciplinaire et Exploratoire and Prématuration-Innovation at Université de Bordeaux, Fondation pour la Recherche et la Biodiversité IDEASHARE & DATASHARE
- 2020 ● Alberta Mennega Stichting fieldwork grant  
1,250€ 📍 Leiden (The Netherlands)
- 2014 ● External mobility grant from the COTE Cluster of Excellence  
3,000€ 📍 Bordeaux (France)
- Research Exchange Program (Inbound)  
AU\$3350 📍 Western Sydney University, NSW (Australia)



## CONFERENCES AND PRESENTATIONS

- 2025 ● Poster - "Weak global trade-off between frost and drought resistance in trees"  
Xylem International Meeting XIM6 📍 Sevilla (Spain)
- 2022 ● Talk - "Trade off in cold and drought tolerance in trees"  
Xylem International Meeting XIM5 📍 Wurzburg (Germany)
- 2017 ● Talk - "Linking changes in gene expression to the macroevolution of flower color in Iochrominae"  
Evolution Meeting 📍 Portland, Oregon
- 2015 ● Talk - "The evolution of cavitation resistance in conifers and the case of world-record *Callitris*"  
Xylem International Meeting XIM2 📍 Bordeaux (France)



## PROFESSIONAL SERVICE

- Guest Lectures (1h)  
MSc courses Plant Physiology, Methods in Biodiversity Analysis 📍 Bordeaux U. & Leiden U.
- Supervision of several 6 month student projects  
1 BSc and 4 MSc - two leading to publications 📍 Bordeaux U. & Leiden U.
- Regular reviewer for scientific journals  
New Phytologist, Tree Physiology, PCE, Evolution, American Journal of Botany, Plos One, BMC Evol. Biol.
- Postdoc and short term contract representative 2021-2022  
Conseil d'Unité UMR Biogeco 📍 Bordeaux (France)